Conditional Matrix Flows for Gaussian Graphical Models

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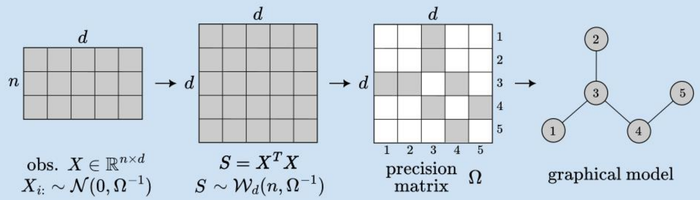
NEURAL INFORMATION PROCESSING SYSTEMS

Gaussian Graphical Models: Graphical Lasso



- GGMs provides **structural** information: $\Omega_{ij} = 0$ implies conditional independence of features i and j given all remaining features
- Maximum likelihood estimate of Ω is not sparse and for $d > n$ is not unique → **penalized likelihood** to encourage sparse Ω

$$\text{argmax}_{\Omega \succ 0}\{\log\det\Omega - \text{Tr}(\tfrac{1}{n}S\Omega) - \lambda\|\Omega\|_0\}$$

Intractable because the objective is highly non-convex

- **Graphical Lasso:** use convex relaxation $\|\Omega\|_0 \longrightarrow \|\Omega\|_1$

Benefits

- **Solution path**: frequentist approach provides solutions as a function of λ

Limitations

- **Over-shrinking:** l_1 norm causes well-known over-shrinking of features
- **No credibility interval**

Bayesian Graphical Lasso

- Bayesian approach: $p(\Omega|S,\lambda) \propto p(S|\Omega)\ p(\Omega|\lambda)$
- Encourage sparsity with double exponential prior (DE)

$$p(S|\Omega) \propto \mathcal{W}_d(n, \Omega^{-1})$$
$$p(\Omega|\lambda) \propto \mathcal{W}_d(d+1, \lambda 1_d)\prod_{i<j}\text{DE}(\omega_{ij}|\lambda)I[\Omega \succ 0]$$

- Recovers the l_1 norm penalized likelihood solution in the MAP limit

Benefits

- **Credibility interval:** can be computed within Bayesian framework

Limitations

- **Expensive posterior sampling:** needs Gibbs sampler (also limits prior choice)
- **Over-shrinking:** prior cannot be generalized to sub-l_1 pseudo-norms
- **Expensive solution path**: independent Markov chains for each λ value

Conditional Matrix Flow for Gaussian Graphical Models

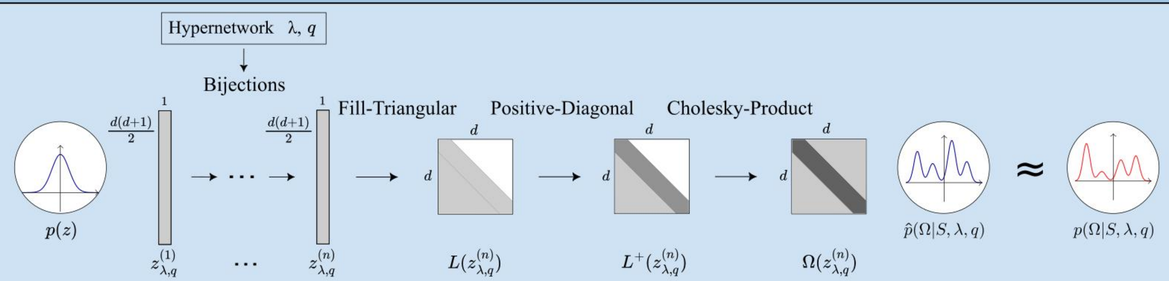
Goal: **unify benefits** of frequentist and Bayesian frameworks while **overcoming limitations**

We propose a general framework for **variational inference** with matrix-variate Normalizing Flows in GGMs. As a key improvement on previous work, we train with one flow a continuum of sparse regression models jointly for all regularization parameters λ and all l_q norms, including non-convex **sub-l_1 pseudo-norms**.

PROPOSED APPROACH

VARIATIONAL INFERENCE	• Posterior approximation and efficient sampling via **Normalizing Flow**: $\mathrm{D_{KL}}\left(\hat{p}(\Omega\|S,\lambda)\|\|p(\Omega\|S,\lambda)\right)$ • Free choice of prior $p(\Omega\|\lambda)$ and likelihood $p(S\|\Omega)$ → model any Bayesian GGMs beyond Lasso
SUB-l_1 PSEUDO-NORMS	• Generalized Gaussian Distributions (GGD) as sparsity-inducing priors corresponding to **l_q (pseudo-) norms** $\quad \text{GGD}(x\|\lambda,q) = \frac{q\lambda^{1/q}}{2\Gamma(1/q)}\exp\{-\lambda\|x\|^q\}$
CONDITIONAL FLOW	• Condition the Normalizing Flow on λ and q to obtain **solution paths** • Can select λ by maximizing marginal likelihood

ARCHITECTURE



- Condition bijections on λ and q and define flow on space of symmetric positive definite matrices **by construction**
- **Efficient log determinant** of Jacobian for Cholesky-Product → $\det \mathcal{J}_{\text{Chol}}(L) = 2^d \prod_{i=1}^{d}(L_{ii})^{d-i+1}$

References

N. Meinshausen and P. Bühlmann (2006). *"High-dimensional graphs and variable selection with the Lasso"* In: Institute of Mathematical Statistics

H. Wang (2012) *"Bayesian Graphical Lasso Models and Efficient Posterior Computation."* In: Bayesian Anal. 7 (4)

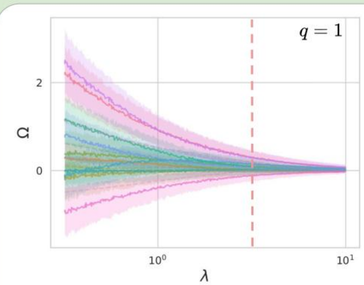
G. Papamakarios et al (2021) *"Normalizing Flows for Probabilistic Modeling and Inference"* In: Journal of Machine Learning Research 22

Combine benefits

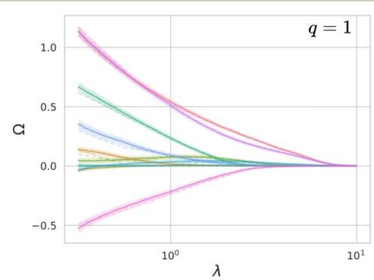
- **Credibility intervals:** sample-based credibility intervals for any λ and q
- **Solution paths**: we recover Bayesian Graphical Lasso ($T = 1$) and Graphical Lasso solution paths ($T \to 0$) by training with simulated annealing

Overcome limitations

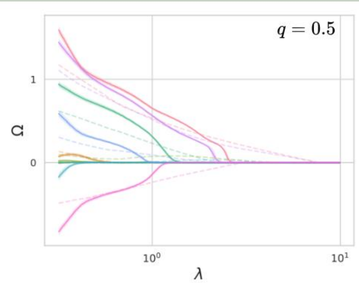
- **Cheap posterior sampling**: no need for sequential Gibbs sampler
- **No over-shrinking:** GGD prior enables exploration of sub-l_1 pseudo-norms
- **Model selection**: direct access to the marginal likelihood for selecting λ



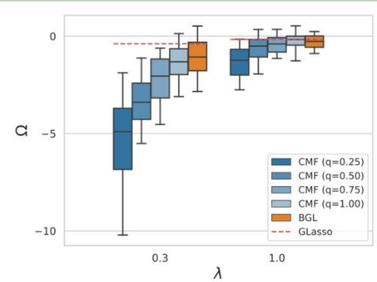
Bayesian solution path at $T = 1$ & **model selection**



Frequentist solution path at $T \to 0$



Solution path in **sub-l_1** regime



Credibility intervals in sub-l_1 regime: **no over-shrinking**